TIVIDY

Class Diagram - final

Matches the final C++11 code

«use»
CommandInvoker
-history_ : vector<shared_ptr<WorkflowAction>>
+executeCommand(cmd : shared_ptr<WorkflowAction>) : bool
+undoLast(workflowId : int) : bool
+getHistory() : vector<string>
invoker 1
history_ 0..*
WorkflowAction
#engine_ : WorkflowEngine&
#workflowId_ : int
#actorId_ : int
+execute() : bool
+undo() : bool
+describe() : string
+getWorkflowId() : int
+getActorId() : int
#target() : shared_ptr<WorkflowInstance>
ApproveCommand
+execute() : bool
+describe() : string
RejectCommand
-reason_ : string
+execute() : bool
+describe() : string
SuspendCommand
+execute() : bool
+undo() : bool
+describe() : string
ResumeCommand
+execute() : bool
+undo() : bool
+describe() : string
CompleteWorkItemCommand
-itemId_ : int
+execute() : bool
+undo() : bool
+describe() : string
EditReportCommand
-text_ : string
-before_ : unique_ptr<ReportMemento>
+execute() : bool
+undo() : bool
+describe() : string
TerminalUI
-facade : WorkflowFacade&
-in : istream&
-out : ostream&
-colour : bool
+TerminalUI(facade : WorkflowFacade&, in : istream&, out : ostream&, colour : bool)
+run() : int
-startWorkflow(presetType : const string&, suggestedRequester : int) : void
-listWorkflows() : void
-openWorkflow(id : int) : void
-showWorkflow(id : int) : void
-createWorkflowType() : void
-showPeople() : void
-showAuditLog() : void
-showHistory(id : int, last : size_t) : void
-ask(question : const string&, suggestion : const string&, refreshOnChange : bool) : string
-askNumber(question : const string&, suggestion : const string&, refreshOnChange : bool) : int
-askPerson(question : const string&, suggestedId : int) : int
-firstWithRole(role : const string&) : int
-addPerson(role : const string&, suggestedName : const string&) : int
-paint(code : const string&, text : const string&) : string
-heading(title : const string&) : void
-ok(text : const string&) : void
-fail(text : const string&) : void
-info(text : const string&) : void
«utility»
IndustrySetup
+configureLabTesting(facade : WorkflowFacade&, timeout : int) : void
+configureLoanApplications(facade : WorkflowFacade&) : void
+configureAllIndustries(facade : WorkflowFacade&, timeout : int) : void
+configureCustomWorkflow(facade, name, description, approverRole, autoApproveSeconds, ...) : void
mediator 1
facade 1
«configure»
WorkflowFacade
-engine : shared_ptr<WorkflowEngine>
-invoker : shared_ptr<CommandInvoker>
-mediator : shared_ptr<WorkflowMediator>
-audit : shared_ptr<AuditLogger>
-approvers : shared_ptr<ApproverObserver>
-definitions : map<string, shared_ptr<const WorkflowDefinition>>
+registerWorkflow(def : shared_ptr<const WorkflowDefinition>) : void
+listWorkflowTypes() : vector<string>
+getDefinition(type : const string&) : shared_ptr<const WorkflowDefinition>
+registerParticipant(p : const Participant&) : void
+listParticipants() : vector<Participant>
+participantName(participantId : int) : string
+submitRequest(type : const string&, request : const WorkflowRequest&) : int
+approveRequest(workflowId : int, actorId : int) : bool
+rejectRequest(workflowId : int, actorId : int, reason : const string&) : bool
+suspendRequest(workflowId : int, actorId : int) : bool
+resumeRequest(workflowId : int, actorId : int) : bool
+completeWorkItem(workflowId : int, itemId : int, actorId : int) : bool
+undoLastAction(workflowId : int) : bool
+canApprove(workflowId : int, actorId : int) : bool
+processTimeouts() : int
+updateReport(workflowId : int, actorId : int, text : const string&) : bool
+getReport(workflowId : int) : string
+getReportRevisions(workflowId : int) : vector<string>
+restoreReport(workflowId : int, index : int, actorId : int) : bool
+listWorkflows() : vector<int>
+getStatus(workflowId : int) : string
+getWorkflow(workflowId : int) : shared_ptr<const WorkflowInstance>
+getAvailableWork(workflowId : int) : vector<shared_ptr<WorkItem>>
+getHistory(workflowId : int) : vector<WorkflowEvent>
+getRequesterInbox(workflowId : int) : vector<string>
+getNotifications(participantId : int) : vector<string>
+getAuditLog() : vector<string>
+getApproverQueue() : vector<string>
+getActionHistory() : vector<string>
-requireInstance(workflowId : int) : shared_ptr<WorkflowInstance>
-run(workflowId : int, cmd : shared_ptr<WorkflowAction>) : bool
engine 1
engine_ 1
engine 1
WorkflowEngine
-instances : map<int, shared_ptr<WorkflowInstance>>
-participants : map<int, Participant>
-globalObservers : vector<shared_ptr<WorkflowObserver>>
-requesterInboxes : map<int, shared_ptr<RequesterObserver>>
-nextId : int
+startWorkflow(def : shared_ptr<const WorkflowDefinition>, req : const WorkflowRequest&) : int
+getInstance(id : int) : shared_ptr<WorkflowInstance>
+listInstances() : vector<int>
+registerParticipant(p : const Participant&) : void
+getParticipant(id : int) : const Participant&
+listParticipants() : vector<Participant>
+addGlobalObserver(o : shared_ptr<WorkflowObserver>) : void
+getRequesterInbox(id : int) : shared_ptr<RequesterObserver>
0..*
Participant
-id_ : int
-name_ : string
-roles_ : vector<string>
-currentWorkload_ : int
-note_ : string
+getId() : int
+getName() : string
+getRoles() : const vector<string>&
+rolesText() : string
+getNote() : string
+hasRole(role : const string&) : bool
+getWorkload() : int
+addWork() : void
+finishWork() : void
WorkflowFactory
+createApprovalPolicy(def : const WorkflowDefinition&) : shared_ptr<ApprovalPolicy>
+createTaskGroup(def : const WorkflowDefinition&) : shared_ptr<CompositeWorkItem>
factory 1
AutoApprovalComponentFactory
+createApprovalPolicy(def : const WorkflowDefinition&) : shared_ptr<ApprovalPolicy>
+createTaskGroup(def : const WorkflowDefinition&) : shared_ptr<CompositeWorkItem>
ManualReviewComponentFactory
+createApprovalPolicy(def : const WorkflowDefinition&) : shared_ptr<ApprovalPolicy>
+createTaskGroup(def : const WorkflowDefinition&) : shared_ptr<CompositeWorkItem>
«create»
root 1
Task
-assignedTo : int
-requiredRole : string
+execute(actorId : int) : bool
+assignTo(participantId : int) : void
+getAssignee() : int
+getRequiredRole() : string
+release() : void
+reopen() : bool
+getChildren() : const vector<shared_ptr<WorkItem>>&
+isComplete() : bool
+isAvailable() : bool
CompositeWorkItem
-children : vector<shared_ptr<WorkItem>>
-sequential : bool
+execute(actorId : int) : bool
+add(item : shared_ptr<WorkItem>) : void
+getChildren() : const vector<shared_ptr<WorkItem>>&
+release() : void
+isComplete() : bool
+createIterator() : unique_ptr<WorkItemIterator>
+createAvailableIterator() : unique_ptr<WorkItemIterator>
+isAvailable() : bool
«create»
WorkItemIterator
+first() : void
+next() : void
+isDone() : bool
+currentItem() : shared_ptr<WorkItem>
DepthFirstIterator
-items : vector<shared_ptr<WorkItem>>
-index : size_t
+DepthFirstIterator(root : const WorkItem&)
-collect(item : const WorkItem&) : void
+first() : void
+next() : void
+isDone() : bool
+currentItem() : shared_ptr<WorkItem>
inner
AvailableWorkIterator
-inner : DepthFirstIterator
+AvailableWorkIterator(root : const WorkItem&)
-skipUnavailable() : void
+first() : void
+next() : void
+isDone() : bool
+currentItem() : shared_ptr<WorkItem>
WorkItem
#id : int
#name : string
#status : string
+WorkItem(id : int, name : const string&, status : const string&)
+execute(actorId : int) : bool
+getId() : int
+getName() : string
+getStatus() : string
+isAvailable() : bool
+isComplete() : bool
+getChildren() : const vector<shared_ptr<WorkItem>>&
+release() : void
+reopen() : bool
+assignTo(participantId : int) : void
+getAssignee() : int
+getRequiredRole() : string
children 0..*
WorkflowDefinition
-name : string
-version : int
-description : string
-approverRole : string
-approvalTimeout : int
-reminderInterval : int
-stages : vector<pair<string, vector<string>>>
-sequentialStages : vector<bool>
-requestFields : vector<pair<string, string>>
-validationChain : shared_ptr<ValidationHandler>
-factory : shared_ptr<WorkflowFactory>
-manualReviewAttribute : string
-manualReviewFrom : double
+setDescription(text : const string&) : WorkflowDefinition&
+setApproverRole(role : const string&) : WorkflowDefinition&
+setApprovalTimeout(seconds : int) : WorkflowDefinition&
+setReminderInterval(seconds : int) : WorkflowDefinition&
+addRequestField(key : const string&, prompt : const string&) : WorkflowDefinition&
+addStage(name : const string&, tasks : const vector<string>&, sequential : bool) : WorkflowDefinition&
+setManualReviewRule(attribute : const string&, from : double) : WorkflowDefinition&
+setValidationChain(chain : shared_ptr<ValidationHandler>) : WorkflowDefinition&
+setFactory(factory : shared_ptr<WorkflowFactory>) : WorkflowDefinition&
+getName() : string
+getVersion() : int
+getStages() : const vector<pair<string, vector<string>>>&
+getValidationChain() : shared_ptr<ValidationHandler>
+getFactory() : shared_ptr<WorkflowFactory>
+getApproverRole() : string
+getApprovalTimeout() : int
+getRequestFields() : const vector<pair<string, string>>&
+isStageSequential(index : size_t) : bool
+getManualReviewAttribute() : string
+getManualReviewFrom() : double
definition 1
WorkflowMediator
+requestValidation(workflowId : int) : void
+validationCompleted(workflowId : int, result : const ValidationResult&) : void
+requestApproval(workflowId : int) : void
+approvalCompleted(workflowId : int, approved : bool) : void
+processTimeouts() : int
+getNotifications(participantId : int) : vector<string>
mediator 1
DefaultWorkflowMediator
-engine : WorkflowEngine&
-validator : ValidationService
-approvals : ApprovalManager
-notifier : NotificationService
+requestValidation(workflowId : int) : void
+validationCompleted(workflowId : int, result : const ValidationResult&) : void
+requestApproval(workflowId : int) : void
+approvalCompleted(workflowId : int, approved : bool) : void
+processTimeouts() : int
+getNotifications(participantId : int) : vector<string>
+getNotifier() : const NotificationService&
ValidationService
+validate(wf : WorkflowInstance&) : void
ApprovalManager
-engine : WorkflowEngine&
+ApprovalManager(engine : WorkflowEngine&)
+requestApproval(wf : WorkflowInstance&) : void
+checkTimeouts() : void
NotificationService
-inbox : map<int, vector<string>>
+send(participantId : int, message : const string&) : void
+getMessages(participantId : int) : vector<string>
+getSentCount() : int
WorkflowColleague
#mediator : WorkflowMediator*
+setMediator(m : WorkflowMediator*) : void
«use»
instances 0..*
WorkflowInstance
-id : int
-definition : shared_ptr<const WorkflowDefinition>
-request : WorkflowRequest
-root : shared_ptr<CompositeWorkItem>
-currentState : unique_ptr<WorkItemState>
-approvalPolicy : shared_ptr<ApprovalPolicy>
-report : WorkflowReport
-history : vector<WorkflowEvent>
-reportHistory : ReportHistory
-participants : map<int, Participant>*
-acceptedAt : steady_clock::time_point
+getId() : int
+getDefinition() : shared_ptr<const WorkflowDefinition>
+getRequest() : const WorkflowRequest&
+getRoot() : shared_ptr<CompositeWorkItem>
+getApprovalPolicy() : shared_ptr<ApprovalPolicy>
+getAcceptedAt() : steady_clock::time_point
+participantName(participantId : int) : string
+validate() : bool
+approve(actorId : int) : bool
+reject(actorId : int, reason : const string&) : bool
+start() : bool
+suspend(actorId : int) : bool
+resume(actorId : int) : bool
+complete() : bool
+completeItem(itemId : int, actorId : int) : bool
+reopenItem(itemId : int, actorId : int) : bool
+getAvailableWork() : vector<shared_ptr<WorkItem>>
+findItem(itemId : int) : shared_ptr<WorkItem>
+allWorkComplete() : bool
+setState(next : WorkItemState*) : void
+getStateName() : string
+canApprove(actor : const Participant&) : bool
+setApprovalPolicy(policy : shared_ptr<ApprovalPolicy>) : void
+shouldAutoApprove() : bool
+addEvent(e : const WorkflowEvent&) : void
+getHistory() : const vector<WorkflowEvent>&
+getReport() : WorkflowReport&
+getReportHistory() : ReportHistory&
*
*
0..*
WorkflowEvent
-type_ : string
-workflowId_ : int
-actorId_ : int
-message_ : string
-timestamp_ : string
+getType() : string
+getWorkflowId() : int
+getActorId() : int
+getMessage() : string
+getTimestamp() : string
+toString() : string
WorkflowSubject
#observers_ : vector<shared_ptr<WorkflowObserver>>
+attach(o : shared_ptr<WorkflowObserver>) : void
+detach(o : shared_ptr<WorkflowObserver>) : void
+notifyObservers(e : const WorkflowEvent&) : void
+getObserverCount() : size_t
0..*
WorkflowObserver
+update(e : const WorkflowEvent&) : void
AuditLogger
-auditLog_ : vector<string>
+update(e : const WorkflowEvent&) : void
+getAuditLog() : const vector<string>&
+printLog() : void
+getLogSize() : size_t
RequesterObserver
-targetRequesterId_ : int
-inbox_ : vector<string>
+update(e : const WorkflowEvent&) : void
+getTargetRequesterId() : int
+getInbox() : const vector<string>&
+getUnreadCount() : size_t
ApproverObserver
-role_ : string
-pendingApprovals_ : vector<string>
+update(e : const WorkflowEvent&) : void
+getRole() : const string&
+getPendingApprovals() : const vector<string>&
+getPendingCount() : size_t
1
WorkflowReport
-content_ : string
-revision_ : int
+edit(text : const string&) : void
+save() : ReportMemento
+restore(m : const ReportMemento&) : void
+getContent() : string
+getRevision() : int
+preview(m : const ReportMemento&, maxLength : size_t) : string
«create»
ReportMemento
-content_ : string
-revision_ : int
+getRevision() : int
«friend» WorkflowReport
snapshots_ 0..*
reportHistory 1
ReportHistory
-snapshots_ : vector<ReportMemento>
+addSnapshot(m : const ReportMemento&) : void
+getSnapshot(index : int) : const ReportMemento&
+size() : int
currentState 1
approvalPolicy 1
WorkItemState
+getStateName() : string
+pending(workflow : WorkflowInstance&) : void
+accepted(workflow : WorkflowInstance&) : void
+rejected(workflow : WorkflowInstance&) : void
+approved(workflow : WorkflowInstance&) : void
+denied(workflow : WorkflowInstance&, message : const string&) : void
+completed(workflow : WorkflowInstance&) : void
+suspended(workflow : WorkflowInstance&) : void
+illegalState(workflow : WorkflowInstance&, newState : const string&) : void
ApprovalPolicy
#approverRole : string
+ApprovalPolicy(approverRole : const string&)
+canApprove(wf : const WorkflowInstance&, actor : const Participant&) : bool
+shouldAutoApprove(wf : const WorkflowInstance&) : bool
+describe() : string
TimeoutApprovalPolicy
-timeoutSeconds : int
+TimeoutApprovalPolicy(approverRole : const string&, seconds : int)
+shouldAutoApprove(wf : const WorkflowInstance&) : bool
+describe() : string
StrictApprovalPolicy
+StrictApprovalPolicy(approverRole : const string&)
+canApprove(wf : const WorkflowInstance&, actor : const Participant&) : bool
+shouldAutoApprove(wf : const WorkflowInstance&) : bool
+describe() : string
PendingState
+getStateName() : string
+accepted(workflow : WorkflowInstance&) : void
+rejected(workflow : WorkflowInstance&) : void
AcceptedState
+getStateName() : string
+approved(workflow : WorkflowInstance&) : void
+denied(workflow : WorkflowInstance&, message : const string&) : void
ApprovedState
+getStateName() : string
+completed(workflow : WorkflowInstance&) : void
+suspended(workflow : WorkflowInstance&) : void
SuspendedState
+getStateName() : string
+accepted(workflow : WorkflowInstance&) : void
RejectedState
+getStateName() : string
DeniedState
+getStateName() : string
CompletedState
+getStateName() : string
WorkflowRequestDirector
-builder : WorkflowBuilder*
+WorkflowRequestDirector(b : WorkflowBuilder*)
+setBuilder(b : WorkflowBuilder*) : void
+constructStandard(input : const map<string, string>&) : void
+constructUrgent(input : const map<string, string>&) : void
builder 1
WorkflowBuilder
+reset() : void
+buildHeader(name : const string&, requesterId : int) : void
+buildAttribute(key : const string&, value : const string&) : void
+buildDeadline(days : int) : void
+buildPriority(level : int) : void
WorkflowRequestBuilder
-request : WorkflowRequest
+reset() : void
+buildHeader(name : const string&, requesterId : int) : void
+buildAttribute(key : const string&, value : const string&) : void
+buildDeadline(days : int) : void
+buildPriority(level : int) : void
+getResult() : WorkflowRequest
RequestSummaryBuilder
-summary : ostringstream
+reset() : void
+buildHeader(name : const string&, requesterId : int) : void
+buildAttribute(key : const string&, value : const string&) : void
+buildDeadline(days : int) : void
+buildPriority(level : int) : void
+getResult() : string
«create»
WorkflowRequest
-name : string
-requesterId : int
-attributes : map<string, string>
-deadlineDays : int
-priority : int
+WorkflowRequest(name : const string&, requesterId : int)
+getName() : string
+getRequesterId() : int
+getAttribute(key : const string&) : string
+setAttribute(key : const string&, value : const string&) : void
+getDeadlineDays() : int
+getPriority() : int
request 1
ValidationHandler
#next : shared_ptr<ValidationHandler>
+add(successor : shared_ptr<ValidationHandler>) : void
+setNext(successor : shared_ptr<ValidationHandler>) : shared_ptr<ValidationHandler>
+canHandle(wf : const WorkflowInstance&) : bool
+handle(wf : const WorkflowInstance&, mode : ChainExecutionMode) : ValidationResult
#check(wf : const WorkflowInstance&) : ValidationResult
#getHandlerName() : string
next 0..1
IdentityVerificationHandler
-directory : shared_ptr<OrganizationDirectory>
#check(wf : const WorkflowInstance&) : ValidationResult
#canHandle(wf : const WorkflowInstance&) : bool
#getHandlerName() : string
AuthorizationHandler
-directory : shared_ptr<OrganizationDirectory>
-activity : string
#check(wf : const WorkflowInstance&) : ValidationResult
#canHandle(wf : const WorkflowInstance&) : bool
#getHandlerName() : string
ThresholdHandler
-attribute : string
-maxValue : double
#check(wf : const WorkflowInstance&) : ValidationResult
#canHandle(wf : const WorkflowInstance&) : bool
#getHandlerName() : string
RequiredAttributesHandler
-keys : vector<string>
#check(wf : const WorkflowInstance&) : ValidationResult
#getHandlerName() : string
«create»
ValidationResult
-valid : bool
-issues : vector<ValidationIssue>
+addIssue(handler : const string&, msg : const string&, sev : IssueSeverity) : void
+merge(other : const ValidationResult&) : void
+isValid() : bool
+getIssues() : const vector<ValidationIssue>&
+str() : string
«struct»
ValidationIssue
+handlerName : string
+message : string
+severity : IssueSeverity
«enumeration»
IssueSeverity
INFO
WARNING
CRITICAL_ERROR
«enumeration»
ChainExecutionMode
FAIL_FAST
AUDIT_ACCUMULATE
directory 1
OrganizationDirectory
+userExists(id : int) : bool
+isAuthorized(id : int, activity : const string&) : bool
RecordSystemAdapter
-legacySystem : shared_ptr<LegacyRecordSystem>
+userExists(id : int) : bool
+isAuthorized(id : int, activity : const string&) : bool
legacySystem 1
LegacyRecordSystem
-legacyDb : map<int, vector<string>>
+queryRecord(recordId : int) : string
+lookupPermissions(recordId : int) : string
+insertRecord(recordId : int, legacyRoles : const vector<string>&) : void

Legend

- 🟧 Facade
- ⬜ Command
- 🟦 Builder
- 🟥 Abstract Factory
- 🟫 Strategy
- 🟩 Composite
- 🟪 Iterator
- 🟦 State
- 🟨 Observer
- 🟪 Chain of Responsibility
- 🟥 Adapter
- 🟩 Mediator
- 🟦 Memento
- ⬜ Core engine / value types
- ⬜ Application layer (client)

Notation

- —▷ generalisation
- ◆— composition (owns)
- ◇— aggregation (shares)
- —> association
- - - -> dependency «create»/«use»

Italic = abstract
WorkflowInstance (yellow) is
multi-role: State & Strategy
Context, Observer Subject,
Command Receiver